MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON
TYPE 8766

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: CW Amplifier, 1.0 kW, 4400 - 5000 MHz, 4 integral cavities, individual tuners, waveguide output, coaxial input, permanent-magnet focusing, forced-air cooling.

OUTLINE AND DIMENSIONS: See Figure 1 herein. WEIGHT: 40 lbs, approx.

ABSOLUTE RATINGS: (Note 1)

Parameter:	Ef	Eb	Ik	Pi	Load	If (surge)	tk	Iby
Unit:	V	kVdc	mAdc	kW	VSWR	a	min.	mAdc
Maximum:	6.8	8.5	600	5.0	2:1	25	---	60
Minimum:	6.0	---	---	---	---	---	3.0	---

Parameter:	Ambient Oper. Pressure	Ambient Temp. Oper.	Ambient Temp. Stg.	Oper. Temp. (Note 2) Collector	Body	Tuner, Fins
Unit:	atmosphere	°C	°C	°C	°C	°C
Maximum:	---	+55	+85	+260	+150	+150
Minimum:	0.64	-30	-65	---	---	---

TEST CONDITIONS:

Ef	Eb	tk	Load	Max. Operating Temp. Collector, Body, Tuner, Fins (Notes 3,4)
V	kVdc	min.	VSWR	
6.5	7.5	3.0	1.05 max.	

CATHODE LEADS: See Figure 1 herein.

RF INPUT CONNECTOR: Mates with Type "N" fitting UG-21D/U or equal.

RF OUTPUT FLANGE: Mates with UG-149A/U flange, or equal

MOUNTING POSITION: Any. (Note 5)

COOLING: Forced air. (Note 4)

FOCUSING: Permanent magnet. (Note 6)

HANDLING CAUTION: (Note 7)

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Method	Requirement or Test	Conditions	AQL (% defec.)	Insp. Level or Code	Symbol	Limits Min	Limits Max	Unit
	GENERAL							
---	Preproduction Sample Approval	Required; (Note 8)	---	---	---	---	---	---
---	Production lot	(Note 9)	---	---	---	---	---	---
---	Marking	(Note 10, 11)	---	---	---	---	---	---
---	Dimensions	(See Fig. 1)	---	---	---	---	---	---
	PREPRODUCTION SAMPLE APPROVAL TESTS, (NOTE 12)							
---	Humidity	(Note 13)	---	---	---	---	---	---
---	Tunable frequency	F = 4400 to 5000 MHz (Note 14)	---	---	---	---	14	turns
	QUALITY CONFORMANCE INSPECTION, PART 1, (NOTE 15)							
---	Visual-mechanical examination	See Fig. 1	---	---	---	---	---	---
1301	Heater current	---	0.65	S3	If	7.4	8.2	A
1256	Cathode current	---	0.65	S3	Ik	450	520	mAdc
1256	Body current	Power output(2)	0.65	S3	Iby	---	35	mAdc
4250	Power output (2)	Tuned for high efficiency F=4400, 4700, and 5,000 MHz Pd = 50 mW (Notes 16, 17)	0.65	S3	Po Po/Pi	1.2 32	--- ---	kW %
4223	Tuner starting torque	t= +20°C, approx.	0.65	S3	---	---	30	oz-in.

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Method	Requirement or Test	Conditions	AQL (% defec.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	QUALITY CONFORMANCE INSPECTION, PART 2							
4271	Bandwidth	-3 db (power level at band limits) Power output (2) conditions except F = 4700 MHz (Note 18)	2.5	S1	ΔF	7	---	MHz
4250	Power output (1)	Tuned for high gain F = 4400, 4700, and 5000 MHz Pd = 10 mW (Notes 16, 17)	6.5	S1	Po	1.0	---	KW
---	Tuner stop torque	t = +20°C, approx. (Note 19)	6.5	S1	---	---	200	oz-in.
	QUALITY CONFORMANCE INSPECTION, PART 3, LIFE							
---	Life test	Group D Power output (2)	---	---	t	1000	---	hrs
	Life test end-point:	Power output (2)			Po	1.0	---	kW
	QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC CHECK, (NOTE 20)							
1042	Shock	No voltages (Note 21)	---	---	---	---	---	---
---	Vibration	No voltages (Note 22)	---	---	---	---	---	---
---	Mechanical tuning fatigue	(Note 23)	---	---	---	100	---	cycles

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NOTES:

1. Refer to Specification MIL-E-1, Section 6 "Note" concerning Absolute Ratings.

2. At any ambient temperature, particular care must be taken (see Note 4, below) to maintain the collector, body, and tuner fins below their maximum rated operating temperatures.

3. The cooling requirements specified or otherwise covered herein (see Note 4, below) shall be effective wherever applicable throughout the testing program herein. The collector temperature shall be measured on the top surface of the collector and the tuner fin temperature shall be measured near the fin shaft. (The thermostat on the collector should be used in an interlock circuit that will remove the beam voltage if the collector temperature becomes abnormally high. This normally closed, Klixon 11041-106-1 or equal, thermostat has a switch rating of 50 Vdc at 2 Amps, or 120 Vac at 1 Amp; it opens at 265.6° ± 10°C (510°F ± 18°F), and closes at 237.8°C ± 10°C (460°F ± 18°F.))

4. Relative to the tube temperature - control requirements specified herein, the following cooling - means accomplishment is recommended. The listed values are approximate, and apply to conditions: sea-level air pressure ≈ 760 mmHg, normal ambient temperature ≈ +20°C.

Air Flow		Inlet Air Pressure	
Collector (ducted) lb/hr	Body (incl. Tuner Fins) (ducted) lb/hr	Collector in.-H_2O	Body in.-H_2O
815	270	2.0	0.75

5. The tube may be operated in any position, but the preferred (particularly for fixed-installation operation) mounting position is vertical, cathode end down. (See Note 11 below.)

6. Ferromagnetic material shall not be brought within 6 inches of the tube; other magnets, not closer than 12 inches.

7. Care should be exercised during all handling for installation, storage, and use of the tube, to prevent impartation of shock or vibration in excess of the values for which the tube is designed.

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8. The Preproduction Sample Approval requirements in U. S. Army Electronics Command Drawing SC-A-46600B hereby replace any Qualification requirements referable to the product covered herein. However, the requirements of paragraphs 3.1.2.1, 3.1.2.2, 3.1.3, and 4.4 in Drawing SC-A-46600B shall be considered not applicable to the product covered herein.

   (Copies of U. S. Army Electronics Command Drawing SC-A-46600B, 19 March 1965, "Preproduction Sample Approval In Lieu Of Qualification Requirements In Specifications For Semiconductor Devices And Electron Tubes" may be obtained from the procuring activity or as directed by the contracting officer.)

9. A production lot is defined as two (2) months production or seventy (70) tubes, whichever is less.

10. Tube-designation marking shall be in accordance with Specification MIL-E-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only.

11. Tube identification and tube serial number shall be marked in the most practical and most readily discernible location on the tube, relative to its normal mounted position (see Note 5, above). Also, indication of the sequential alignment of the tuning cavities shall be effected by marking, equivalent to that shown in Fig. 1 herein.

12. Although all tests covered herein are required to be performed during Preproduction Sample Approval inspection, the 2 tests listed under this caption are performed, normally, during Preproduction Sample Approval only. A failure of any one tube in any of these tests shall be cause for decision of preproduction-sample disapproval.

13. The electron tube shall be subjected to a relative humidity of 90 to 95 percent for 96 hours in accordance with Standard MIL-STD-202, Method 103, Test Condition B. Upon completion of this test, the tube shall meet the requirements for Cathode Current and Power Output (2) herein.

14. The electron tube shall be capable of being tuned smoothly and without discontinuities over the frequency range including the limits specified. The number of turns of the tuner required to tune the tube over the frequency range shall be as specified herein. From lowest to highest frequency, tuning shall be in a clockwise direction.

15. The AQL for the combined defectives for attributes in Quality Conformance Inspection, Part 1, exclusive of "Visual-Mechanical examination", shall be 1.0%. A tube having more than one defect shall be counted as one defective.

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16. With tube synchronously tuned, cavity # 3 shall be set approximately one turn high in frequency. Drive power shall be increased to indicated drive power, and then cavity # 3 shall be retuned to maximum power output. Minor adjustments of other cavites may be made in order to maximize power output.

17. The minimum power output condition for Power Output (1) and (2) shall be available continuously without exceeding the maximum operating temperatures specified under "ABSOLUTE RATINGS" herein, while providing air cooling with an ambient inlet temperature, at sea level, of 45°C - 55°C, and an approximate air flow of 380 lb/hr against a back pressure of 0.45 inches of water, maximum.

18. A frequency swept driver shall be used to view the power output response during the bandwidth test. (See Notes 16 and 17 above.)

19. The tuning mechanism of each individual cavity shall withstand the maximum specified stop torque at the extreme ends of the tuning range, without damage.

20. For periodic tests, sample tubes are to be selected at random intervals so that during any twelve month period a minimum of two tubes shall have been successfully tested. In the event of failure of one or more tubes in a sample, these tests are to become a part of Quality Conformance Inspection, part 2, until two successive lots have been successfully completed on these tests. After two successive lots have thus been successfully completed, the periodic tests shall then be eligible to be resumed on a twelve month basis.

21. The tube shall be subjected (non-operating) to shock test in accordance with MIL-STD-202, Test Method 213, Test Condition A except:

    Peak value (g's) = 20

    Velocity change (VI) = 140.01

    After this test, the tube(s) shall meet the Power Output (2) and Cathode Current requirements herein.

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22. The tube shall be mounted rigidily on the vibration table, and subjected to vibration, as follows:

    a. The tube shall be vibrated under simple harmonic motion through 10 complete scan cycles in each of 3 mutually perpendicular planes, one of which shall be parallel to the beam axis. Each scan cycle shall be of ten minutes duration consisting of a continously varying vibration frequency between the limits of 10 to 500 to 10 Hz. The vibration frequency shall be divided into 3 steps with peak acceleration as shown below:

        G = 10, between 10 and 100 Hz, except a limited displacement of 0.06-inch double amplitude.

        G = 5, between 100 and 150 Hz.

        G = 2, between 150 and 500 Hz.

    b. The waveguide and collector may be supported by a suitable fixture or fixtures such that, when the tube is subject to vibration, the ratio of the peak acceleration, -as measured on the flange of the waveguide or on the top center of the collector, with respect to the peak acceleration on the base plate, has a minimum value of 1:1.

    c. It is also permissible to secure the tuner shafts so as to prevent rotation during the vibration test.

    After this test, the tube(s) shall meet the Power Output (2) requirement therein.

23. The tube shall be operable after each tuning mechanism is subjected to the tuning cycles specified, from one extreme end of the tuning range to the other and back again.

24. Each electron tube shall be furnished with dust covers for protection of input and output connector elements.

25. Unless otherwise specified in contract or order, each tube shall, as a minimum protection against moisture and moisture vapor, be sealed within a moisture-vaporproof barrier bag, or equivalent, normally acceptable under packaging requirements of Specification MIL-E-75.

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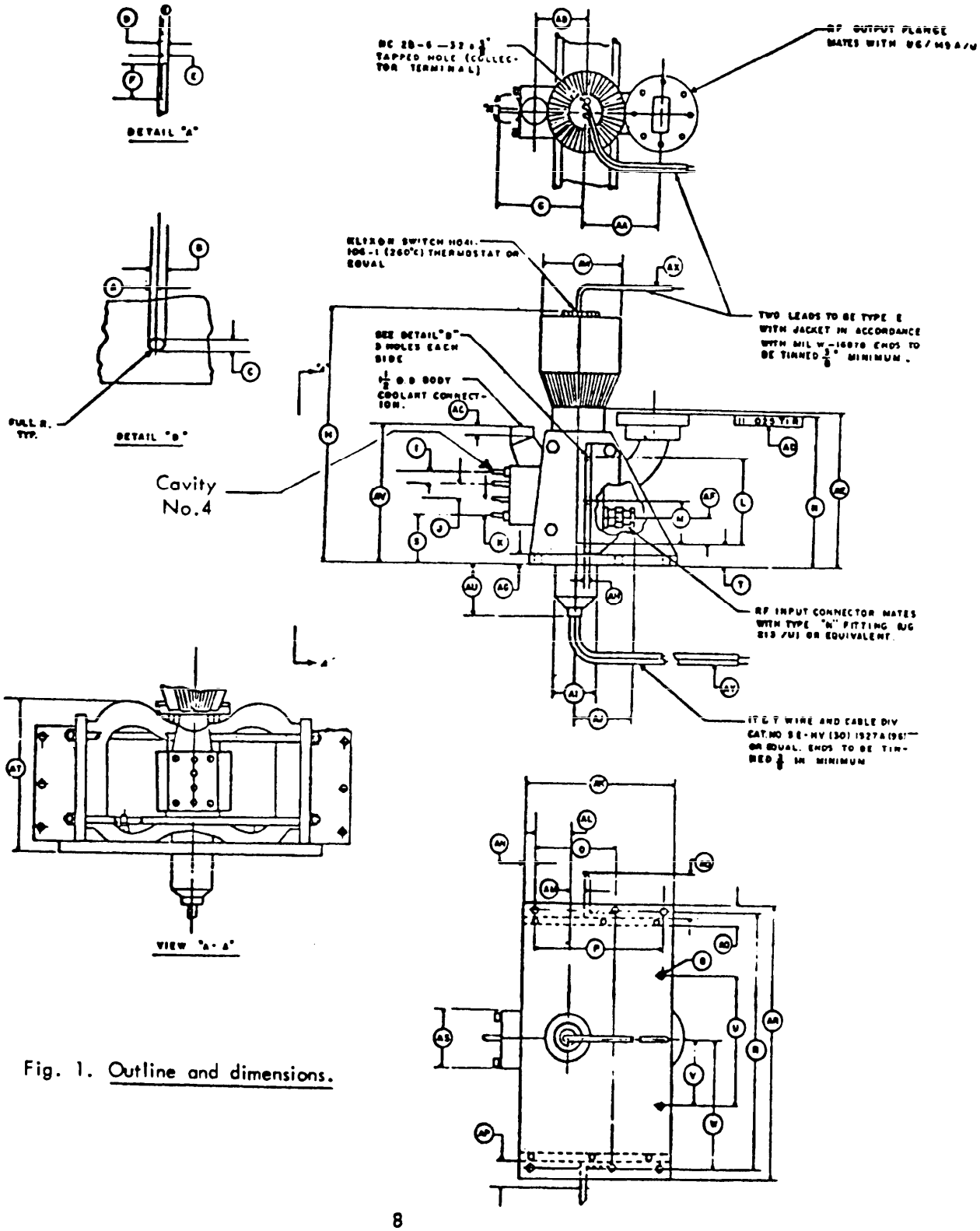


Fig. 1. Outline and dimensions.

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Ltr	Minimum	Maximum
	Qual. Conf. Insp., Part 2 6.5 AQL, Insp. Level S-1	
A	.141 (3.58)	.171 (4.34)
B	.297 (7.54)	.327 (8.30)
C	.235 (5.96)	.265 (6.73)
D	.247 (6.27)	.249 (6.32)
E	.170 (4.31)	.180 (4.57)
F	.740 (18.79)	.760 (1930)
G	4.481 (113.81)	4.605 (116.96)
H	12.875 (327.02)	13.125 (333.37)
I	.669 (16.99)	.731 (18.56)
J	.669 (16.99)	.731 (18.56)
K	.669 (16.99)	.731 (18.56)
L	4.735 (120.26)	4.763 (120.98)
M	2.360 (59.94)	2.390 (60.70)
N	7.375 (187.32)	7.625 (193.67)
O	3.990 (101.34)	4.010 (101.85)
P	6.865 (174.37)	6.885 (174.87)
Q	.297 (7.54)	.327 (8.30)
R	12.090 (307.08)	13.010 (330.45)
S	2.375 (60.32)	2.625 (66.67)
T	.938 (23.82)	1.062 (26.97)
U	6.490 (164.84)	6.510 (165.35)
V	3.235 (82.16)	3.265 (82.93)
W	6.485 (164.71)	6.515 (165.48)

Ltr	Minimum	Maximum
	Qual. Conf. Insp., Part 3 (See Note 20 herein)	
AA	3.968(100.78)	4.032(102.41)
AB	2.875 (73.02)	3.125 (79.37)
AC	.485 (12.31)	.515 (13.08)
AD	---	.025 (.635)
AE	7.250 (184.15)	7.750 (196.85)
AF	2.375 (60.32)	2.625 (66.67)
AG	.485 (12.31)	.515 (13.08)
AH	.172 (4.36)	.202 (5.13)
AI	2.315 (58.80)	2.375 (60.32)
AJ	2.900 (73.66)	3.020 (76.70)
AK	7.860 (199.64)	7.890 (200.40)
AL	1.985 (50.41)	2.015 (51.18)
AM	.485 (12.31)	.515 (13.08)
AN	.485 (12.31)	.515 (13.08)
AO	1.985 (50.41)	2.015 (51.18)
AP	1.422 (36.11)	1.452 (36.88)
AQ	.875 (22.22)	.999 (25.37)
AR	13.735 (348.86)	13.765 (349.63)
AS	2.540 (64.51)	2.600 (66.04)
AT	7.375 (187.32)	7.625 (193.67)
AU	2.375 (60.32)	2.625 (66.67)
AV	6.875 (174.62)	7.000 (177.80)
AW	4.063 (103.20)	4.187 (106.34)
AX	18.0 (457.20)	---
AY	18.0 (457.20)	---

NOTE:

1. Dimensions in inches with metric equivalents (mm) in parentheses.

Figure 1 - (Cont'd).- Outline and dimensions.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/1669(EL) ELECTRON TUBE, KLYSTRON TYPE 8766

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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